Saskia Frisby
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Employment

2024–2027 Postdoctoral Research Associate (MRC Cognition & Brain Sciences Unit)

- Characterised the role of frequency bands in representing multidimensional semantic information via time-frequency analysis of electrocorticography (ECoG) data and state-of-the art machine learning methods.
- Developed methods primer to bring novel decoding methods (including Representational Similarity Learning and Sparse-Overlapping-Sets LASSO) to a broad, interdisciplinary audience.

Education

2020–2024 PhD in Medical Sciences (MRC Cognition and Brain Sciences Unit)
Title: "A comparison of imaging modalities and decoding methods for detecting semantic information in the brain".

- Produced theoretical review that unifies contradictory theories of semantic cognition and provides a field-guide to decoding of neuroimaging data.
- Identified parallels between representation of semantic information in time-frequency decomposed ECoG data and in computational models.
- Optimised novel 7T-fMRI protocol that will enable imaging of the anterior temporal lobe in both healthy participants and patients.
- Adjudicated 12 competing hypotheses about semantic representation using a comparative multivariate analysis applied to 7T-fMRI data.

2017–2020 BA (Hons) Psychological and Behavioural Sciences, Gonville & Caius College, University of Cambridge
First class in all three years.

Visiting positions

June–July 2023 University of Wisconsin-Madison: Visiting PhD Student in the Knowledge and Concepts Lab led by Professor Tim Rogers.

- Contributed to the development of WISC MVPA (a toolbox for decoding with novel methods including Representational Similarity Learning and Sparse-Overlapping-Sets LASSO) into a publicly available resource.

Funding

2026	Cambridge Language Sciences Incubator Fund (£4,884; PI)
2025	Enhancing Research Culture Funding (£13,500; PI)
2025	University of Cambridge Research Culture Conference Funding (£138)
2020–2024	Stanley Elmore Studentship in the Biomedical Sciences, Gonville & Caius College, Cambridge (£6000 p.a.; stipendiary "top-up" award for academic excellence)
2020–2024	Medical Research Council Studentship (stipend plus all university fees)

2023	Gonville & Caius Travel and Research Expenses Grant (£850)
2022	Experimental Psychology Society Study Visit Grant (£3500)
2019	Summer Vacation Project Grant from G. C. Grindley Fund (£1200)
2019	Gonville & Caius Summer Internship Grant (£400)

Academic awards

2024 & 2026	ARRC Research Culture Celebration Nominee (University-level recognition for positive contribution to research culture)
2022	Prize for best poster, Cambridge Imaging Festival
2020	Departmental Commendation, Department of Psychology, University of Cambridge (awarded to 5 best-performing third year Psychology students)
2019	Senior Scholarship, Gonville & Caius College, Cambridge (for first in second year)
2018	Scholarship, Gonville & Caius College, Cambridge (for first in first year)

Publications

<u>Frisby, S. L.</u>, Halai, A. D., Cox, C. R., Lambon Ralph, M. A., & Rogers, T. T. (2023). Decoding semantic representations in mind and brain. *Trends in Cognitive Sciences, 27* (3), pp. 27-53.

<u>Frisby, S. L.</u>, Halai, A. D., Cox, C. R., Clarke, A., Shimotake, A., Kikuchi, T., Kuneida, T., Arakawa, Y., Takahashi, R., Matsumoto, R., Ikeda, A., Rogers, T. T., & Lambon Ralph, M. A. (2026). All spectral frequencies of neural activity reveal semantic representation in the human anterior ventral temporal cortex. *Imaging Neuroscience.* Preprint: https://doi.org/10.1162/IMAG.a.1201

Papers under review

<u>Frisby, S. L.</u>, Correia, M. M., Zhang, M., Rodgers, C. T., Rogers, T. T., Lambon Ralph, M. A., & Halai, A. D. (2025). Optimising 7T-fMRI for imaging regions of magnetic susceptibility. *Human Brain Mapping*. Preprint: https://doi.org/10.1101/2025.03.17.643748

Preprints

<u>Frisby, S. L.</u>, Cox, C. R., Halai, A. D., Lambon Ralph, M. A., & Rogers, T. T. (2025). Comparative multivariate decoding adjudicates theories of semantic representation in the anterior temporal lobes and the rest of the cortex.
Preprint: https://doi.org/10.1101/2025.08.22.671718

Papers in preparation

Shimotake, A., <u>Frisby, S. L.</u>, Kikuchi, T., Ikeda, A., Lambon Ralph, M. A., & Matsumoto, R. Elucidating human brain function and pathology using electrocorticography. *Psychiatry and Clinical Neuroscience* (invited review).

Cox, C. R., Rogers, T.T., & <u>Frisby, S. L.</u> Neural decoding with regularised regression: a framework for investigating the nature of representation.

<u>Frisby, S. L.</u>, Cox, C. R., Halai, A. D., Shimotake, A., Kikuchi, T., Kuneida, T., Arakawa, Y., Takahashi, R., Matsumoto, R., Ikeda, A., Rogers, T. T., & Lambon Ralph, M. A. Multidimensional semantic information in the ventral anterior temporal cortex is encoded conjointly across frequency bands.

Invited talks

May 2025	"All spectral frequencies of neural activity reveal semantic representation in the human anterior ventral temporal cortex" – Waseda University, Japan.
May 2025	"Decoding semantic representations in mind and brain" – Georgia Tech, USA (online).
April 2025	"Decoding semantics with 7T-fMRI: Convergent evidence and divergent discovery" – Bangor University.
Jan 2025	"Research culture: one early career researcher's perspective" (panel discussion), Cambridge University Annual Meeting for Directors of Postgraduate Education.
Nov 2023	"Research culture: one early career researcher's perspective" (talk and panel discussion), Cambridge University Science and Policy Exchange.
July 2023	"A comparison of imaging modalities and decoding methodologies for detecting semantic information" – Department of Psychology, Louisiana State University, USA (online).

Invited conference presentations

August 2025	"Multivariate decoding reconciles neuroimaging with computational modelling" – CogSci, San Francisco.

Talks

Nov 2023	"Representational Similarity Learning" – MRC Cognition and Brain Sciences Unit Methods Day.
Nov 2023	"ECoG in Japan and Cambridge: Past and current results" – Kobe University, Japan (online).
June 2023	"Decoding semantic representations in mind and brain" – AI + Society Seminar, University of Wisconsin-Madison, USA.
Dec 2022	"Intracranial adventures: Using electrocorticography (ECoG) to characterise semantic processes" – MRC Cognition and Brain Sciences Unit Methods Day.
May 2022	"Decoding contemporary approaches to semantic representations in cortex" – MRC Cognition and Brain Sciences Unit Wednesday Lunchtime Seminar.
Nov 2021	"What kind of neural code underpins semantic representations?" – Mental Sciences Club, Department of Philosophy, University of Cambridge.

Conference presentations

October 2024	Annual Meeting of the Society for the Neurobiology of Language – presented poster entitled "Ultra-high-field (7T) fMRI reveals graded semantic structure in the ventral anterior temporal cortex".
October 2023	Annual Meeting of the Society for the Neurobiology of Language – presented poster entitled "Optimising 7T-fMRI for imaging the anterior temporal lobe".
July 2023	Annual Meeting of the Organization for Human Brain Mapping – presented poster entitled "Human grid electrode ECoG reveals a cross-frequency semantic code in anterior temporal cortex".

May 2023	Spring Meeting of the British Neuropsychological Society – presented poster entitled "Human grid electrode ECoG reveals a cross-frequency semantic code in anterior temporal cortex".
October 2022	Annual Meeting of the Society for the Neurobiology of Language – presented poster entitled "Human grid electrode ECoG reveals that local neural activity in anterior temporal cortex expresses semantic information."
June 2022	Cambridge Imaging Festival – presented poster, and gave Slide Slam, entitled "Multi-echo, but not parallel transmit or multiband, improves imaging of semantic cognition with 7T-fMRI".

Programming skills

- MATLAB: development of toolboxes implementing novel machine learning methods (WISC MVPA), decoding with regularised logistic regression (with glmnet), development of a novel intracranial data preprocessing pipeline (with eeglab, CleanLine, and SASICA), conversion of data to BIDS format to facilitate data sharing, fitting general linear models (with SPM12) to 7T-fMRI data, testing for slice leakage in 7T-fMRI data, statistical analysis of behavioural data, data visualisation.
- Bash: development of novel 7T-fMRI data preprocessing pipeline (with AFNI, FSL, tedana, and ANTS), conversion of data to BIDS format (with heudiconv), animating figures to visualise change in decoding performance over time (with ffmpeg).
- Python: plotting decoding coefficients on the cortical surface (with nilearn).
- R: decoding with regularised logistic regression (with glmnet), data visualisation.
- Slurm: Optimising computationally-demanding workflows for the high-performance computing (HPC) cluster at the MRC Cognition & Brain Sciences Unit.
- HTCondor: Optimising computationally-demanding workflows for high-throughput computing systems including the Centre for High-Throughput Computing (University of Wisconsin-Madison) and the Open Science Grid (USA-wide).

Technical skills

- Collecting 7T-fMRI data: recruiting participants, collaborating with MRI physicists to select appropriate acquisition parameters, and working with radiographers to ensure that data are consistently of high quality.
- Decoding 7T-fMRI and ECoG data: theoretical and practical understanding of standard decoding methods, including regression with LASSO regularisation, and novel approaches, including regression with Sparse-Overlapping-Sets LASSO regularisation and Representational Similarity Learning with group-ordered-weighted LASSO regularisation.

Collaborations

- Dr Christopher Cox – Department of Psychology, Louisiana State University, USA
- Professor Rob Nowak – Department of Electrical and Computer Engineering, University of Wisconsin-Madison, USA
- Professor Riki Matsumoto – Department of Neurology, Kyoto University, Japan
- Dr Akihiro Shimotake – Department of Neurology, Kyoto University, Japan
- Dr Alex Clarke – Department of Psychology, University of Warwick

- Tim Dick – Maastricht Centre for Systems Biology and Bioinformatics, University of Maastricht, Netherlands
- Ivette Colón – Department of Psychology, McGill University, Canada
- Liz Simmonds – Head of Research Culture, University of Cambridge, UK

Supervision

Sept–Dec 2025 Rudolf Kirchner (intern; next destination: Research Assistant, MRC Cognition and Brain Sciences Unit; thereafter, PhD, MRC Cognition & Brain Sciences Unit, funded by the Doctoral Training Programme in Medical Research)

Teaching

Nov 2025–present Lecturer in Improving Scientific Practices in Cognitive Neuroscience, MPhil in Cognitive Neuroscience, MRC Cognition & Brain Sciences Unit, University of Cambridge.

- Introduced students to intracranial electrophysiology, multivariate decoding, and neural network modelling for the first time.
- Developed students' ability to critically analyse study design.
- Received excellent feedback – students stated that the clear explanations given in this seminar provided a strong foundation for teaching provided later in the course.

Nov 2025–present Chair, Research Culture Wednesday Lunchtime Seminar, MRC Cognition & Brain Sciences Unit, University of Cambridge.

- Designed a syllabus of five research culture seminars, designed to educate department members at all levels (PIs, fellows, postdocs, graduate students, research assistants, professional services staff) about issues in research culture on which they, as individuals, can make an important and immediate difference
- Convened local and national research culture experts to provide lectures.
- Received feedback that these seminars have catalysed improvements in research practice.

2022–2024 Supervisor for PBS2: Psychological Enquiry and Methods (16 weeks), part of 1st year Psychological and Behavioural Sciences, Robinson College (2022–2024) and St John's College (2023–2024), University of Cambridge.

- Designed and delivered personalised seminars to groups of 1–4 students.
- Set and marked essays, providing supportive and constructive written feedback.
- Managed challenging student behaviour in a thoughtful and sensitive manner.

Nov 2021–present Lecturer in Research Culture, part of the Robust Behavioural Science Course offered to MPhil and first-year PhD students, MRC Cognition and Brain Sciences Unit and Departments of Psychology and Psychiatry, University of Cambridge.

- Designed an interactive seminar for ~30 students to promote their understanding of University and departmental research culture strategy.
- Facilitated discussion, encouraging students to consider their own role in improving research culture.

Oct 2025 Lecturer in Research Culture, part of the induction course for MPhil in Population Health Sciences, Department of Public Health and Primary Care.

- Adapted the interactive seminar that I designed for the MRC Cognition & Brain Sciences Unit to meet the specific needs of a new course and a new department.

Leadership

2021–present MRC Cognition and Brain Sciences Unit Working Group on Research Culture (founder and group lead)

- Established group to address areas for departmental culture improvement.
- Acquired funding (£13,500) for the group's activities.
- Oversaw detailed surveys of Unit culture to assess individuals' concerns and identify areas for improvement.
- Advocated successfully for the inclusion of research culture teaching in the MRC Cognition and Brain Science's Unit's teaching programme for MPhil and PhD students.
- Introduced the Research Culture Wednesday Lunchtime Seminar to provide research culture teaching for all members of the department.
- Supervised students writing a research culture bulletin to celebrate research culture activities taking place within the department.
- Coordinated efforts with existing departmental committees (Open Science; Equality, Diversity and Inclusion), with research culture groups in other departments (Psychology, Psychiatry, Public Health and Primary Care, Sociology, Chemistry, Astronomy), and with University research culture initiatives.
- Represented the department at two conferences: Warwick International Research Culture Conference and the Clinical School People, Culture, and Environment Conference.
- Planned a University-wide Research Culture Conference to be held at the department in December 2026.

2020–present MRC Cognition and Brain Sciences Unit Open Science Committee (member)

- Influenced decision-making about departmental open science policy surrounding teaching, hiring, publishing, and relationships with other open science groups in the University, the UK and beyond.

Outreach

Aug 2022 "Imaging the anterior temporal lobe with fMRI" – presentation to 20 prospective Psychology applicants from disadvantaged backgrounds as part of the Sutton Trust Summer School.

May 2022 "All about the brain" – interactive workshop, delivered via Zoom, for drama group members aged 11-13 at the Storyhouse Theatre, Chester.

March 2022 "What kind of neural code underpins semantic representations?" – pre-recorded talk for general public interest at the Cambridge Festival.

Ad-hoc reviewing

- Imaging Neuroscience
- PLoS Biology
- iScience
- Journal of Cognitive Neuroscience
- Brain and Language
- Language, Cognition & Neuroscience

Professional memberships

- Cognitive Science Society – Member
- Experimental Psychology Society – Postgraduate Member
- Society for the Neurobiology of Language – Student Member
- Organization for Human Brain Mapping – Student Member
- British Neuropsychological Society – Associate Member

Referees

- Professor Matt Lambon Ralph (current line manager, PhD supervisor) – matt.lambon-ralph@mrc-cbu.cam.ac.uk
- Professor Tim Rogers (PhD supervisor) – ttrogers@wisc.edu